Lab Tasks

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Section: A

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Course: Artificial Intelligence Lab (CS - 325L)

Program: (BS) - Computer Science

Lab #01

Activity- 1:

Write a script that take user input for a number then adds 3 to that number. Then multiplies the result by 2, subtract 4, then again adds 3, then print the result.

```
num = int(input("Enter a number:"))
result=(((num+3)*2)-4)+3
print("Final result:",result)
```

```
Enter a number: 2
Final result: 9
```

Activity- 2:

Write a script that takes input as radius then calculate area of circle. (Hint: A= πr^2).

```
import math
radius=float(input("Enter the radius:"))
area=math.pi*radius**2
print("Area of the circle:",area)
```

```
Enter the radius: 3.2
Area of the circle: 32.169908772759484
```

Activity- 3:

Write a Python script that asks users for their favourite color. Create the following output (assuming blue is the chosen color) (hint: use „+‟ and „*‟)

blueblueblueblueblueblueblueblueblueblue
blue   blue
blueblueblueblueblueblueblueblueblueblue

```
color=input("Enter your favourite color:")
print(color*10)
print(color+ "                              " +color)
print(color*10)
```

```
Enter your favourite color: blue
blueblueblueblueblueblueblueblueblueblue
blue                                blue
blueblueblueblueblueblueblueblueblueblue
```

Activity- 4:

Store a person‟s name, and include some „*‟ characters at the beginning and end of the name. Print the name once, so the „*‟ around the name is displayed. Then print the name using each of the three stripping functions, lstrip(), rstrip(), and strip().

```
name= "***Faisal***"
print("Original:",name)
print("lstrip():",name.lstrip('*'))
print("rstrip():",name.rstrip('*'))
print("strip():",name.strip('*'))
```

```
Original: ***Faisal***
lstrip(): Faisal***
rstrip(): ***Faisal
strip(): Faisal
```

Activity- 5:
Write a function called absolute_num() that accepts one parameter, num. The function should return only positive value, and apply condition on it. This function returns the absolute value of the entered number.

```
def absolute_num(num):
    if num<0:
        return -num
    else:
        return num

print(absolute_num(int(input("Enter a number:"))))
```

```
Enter a number: -5
5
```

Activity- 6:
Write a function called describe_city( ) that accepts the name of a city and its country. The function should print a simple sentence, such as Karachi is in Pakistan. Give the parameter for the country a default value. Call your function for three different cities, at least one of which is in the default country.

```
def describe_city(city,country="Pakistan"):
    print(f"{city} is in {country}.")

describe_city("Karachi")
describe_city("Lahore")
describe_city("Tokyo","Japan")
```

```
Karachi is in Pakistan.
Lahore is in Pakistan.
Tokyo is in Japan.
```

Activity- 7:
Write a python script that take a user input and to create the multiplication table (from 1 to 10) of that number.

```
num=int(input("Enter a number for multipication table:"))
for i in range(1,11):
    print(f"{num}x{i} = {num*i}")
```

```
Enter a number for multipication table: 3
3x1 = 3
3x2 = 6
3x3 = 9
3x4 = 12
3x5 = 15
3x6 = 18
3x7 = 21
3x8 = 24
3x9 = 27
3x10 = 30
```

Activity- 8:
Write a Python program that prints all the numbers from 0 to 6 except 3 and 6.
Note: Use 'continue' statement.

```
for i in range(7):
    if i == 3 or i==6:
        continue
    print(i)
```

```
0
1
2
4
5
```

Activity- 9:
Stages of Life: Write an if-elif-else chain that determines a person‟s stage of life. Set a value for the variable age, and then:

- **If the person is less than 2 years old, print a message that the person is a baby.**
- **If the person is at least 4 years old but less than 13, print a message that the person is a kid.**
- **If the person is at least 13 years old but less than 20, print a message that the person is a teenager.**
- **If the person is at least 20 years old but less than 65, print a message that the person is an adult.**
- **If the person is age 65 or older, print a message that the person is an old.**

```
age = int(input("Enter your age: "))

if age < 2:
    print("The person is a baby.")
elif age >= 4 and age < 13:
    print("The person is a kid.")
elif age >= 13 and age < 20:
    print("The person is a teenager.")
elif age >= 20 and age < 65:
    print("The person is an adult.")
elif age >= 65:
    print("The person is old.")
else:
    print("The person is a toddler.")
```

```
Enter your age:  22
The person is an adult.
```